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PERI-OPERATIVE

Peri-operative Management of patients on Antiplatelets & Anticoagulants

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Key Points

It is important to consider the timing of anticoagulant administration before and after surgery by taking into account the thrombotic risk of the patient and bleeding risk from surgery, neuraxial interventions and nerve blocks in order to minimise thrombotic and bleeding risk.

There are many new anticoagulant therapies in current use. Eg. Direct-acting oral anticoagulants (DOACs) have emerged as an alternative to vitamin K antagonists for prevention of stroke in patients with non-valvular atrial fibrillation and venous thrombosis.

Neuraxial interventions carry a rare, but potentially serious risk of vertebral canal haematoma, particularly in patients on anticoagulant therapy.

For Patients undergoing NECK OF FEMUR FIXATION Surgery go to page 5-6

Introduction

This document helps to guide clinicians in minimising both the thrombotic risk and the bleeding risk from surgical and anaesthetic interventions for patients on anti-coagulant therapy.

This guidance should not be taken as being prohibitive or indicative. An individual patient's circumstances must be considered. For example, there may be situations when although the use of a regional technique for a patient with abnormal coagulation may put the patient at significant risk, the alternative (a general anaesthetic) may expose them to even greater risk.

Where possible, patients should be given all the information they need to make an informed choice.

For patient undergoing fractured Neck of Femur fixation surgery, refer to AAGBI M* of Hip Fractures 2020.

Evaluating the Risk of Bleeding

Interventions considered low bleeding risk:

- Endoscopy with biopsy, prostate or bladder biopsy, minor orthopaedic surgery, pacemaker, angiography.
- Superficial plexus blocks and peripheral nerve blocks at compressible sites[1].

Interventions considered high bleeding risk:

- Major orthopaedic, abdominal, breast, airway, vascular surgery, major urology, liver or kidney biopsy.
- Complex endoscopy (e.g. polypectomy, ERCP with sphincterotomy etc.)
- Neuraxial procedures, (including removal of neuraxial catheter), Deep plexus blocks and peripheral nerve blocks at non-compressible sites[1].

[1]It is recommended that the same guidelines on antiplatelets and anticoagulants for neuraxial injections be followed for peripheral nerve blocks. However, a risk benefit judgement can be made for superfical plexus blocks or those at compressible sites eg Axillary brachial plexus blocks.

Evaluating the Risk of Thrombosis: To Bridge or not to Bridge

DOACs

Pre-operative

- Pre-operative bridging with therapeutic Low Molecular Weight Heparin (LMWH) is not required for patients on a DOAC. However, it may be considered in patients with high risk of thrombosis e.g. VTE within 3 months or off licence use in coronary artery stenting (discuss with Cardiology).

Post-operative

- Post-operative bridging with prophylactic doses of LMWH may be considered if the patient is unable to recommence the DOAC within 24 hours of surgery due to:
  - high bleeding risk procedure: Do not re-introduce DOAC at full dose until at least 48 h post-procedure.
  - inability to take oral therapy.
- Never administer LMWH in conjunction with DOAC therapy.

Warfarin and other Vitamin K antagonists

Pre-operative

- Bridging with therapeutic LMWH may be required for patients on Warfarin if they have:
  - Very high thromboembolic risk (see Fig.1).
  - Bileaflet aortic valve with risk factors for thromboembolism (see Fig.1).

- If bridging is required, then:
  - Warfarin is stopped for 5 days before the procedure (5th day being the day of the procedure).
  - The last dose of therapeutic LMWH should be at least 24 hours before surgery and if on a once a day regimen, some recommend the last dose is halved for high risk surgery.

Fig .1 – Thromboembolic Risk Stratification

Risk of Thromboembolism			
Indication	**Very High**	**High**	**Moderate**
Atrial Fibrillation (AF)	Stroke or TIA within 3 months $CHADS_2$ Score >4 Rheumatic mitral stenosis and AF	$CHADS_2$ Score of 3 or 4 Stroke or TIA > 3 months prior	$CHADS_2$ Score ≤ 2 (and no prior history of Stroke or TIA)
Metallic Heart Valve*	Mitral valve prosthesis Cage-ball or tilting disc aortic valve prosthesis Stroke or TIA within 6 months	Bileaflet aortic valve with risk factors for valve thrombosis / embolic stroke**	Bileaflet aortic valve with no other risk factors
Venous Thromboembolism (VTE)	VTE within 3 months Severe thrombophilia*** VTE with INR target of 3.5	VTE 3 – 12 months prior Non-severe thrombophilia****	VTE > 12 months prior and no other risk factors

* All DOACs are currently contraindicated for patients with a metallic heart valve

**AF, prior CVA / TIA, hypertension, diabetes mellitus, congestive cardiac failure, age > 75 years

*** Active cancer, antiphospholipid syndrome, deficiency of protein C, protein S, or Antithrombin, multiple thrombophilia

**** Heterozygous for Factor V Leiden or prothrombin gene mutation

Patients with atrial fibrillation who have a $CHADS_2$ score of ≤4 and who have not had a stroke or transient ischaemic attack (TIA) in last three months should not receive bridging.

Patients with a bi-leaflet aortic mechanical heart valve (MHV) with no other risk factors (AF, prior stroke/TIA, Hypertension, Diabetes, CCF or age ≥75 years) do not require bridging whilst it should be considered in all other MHV patients.

Post-operative

- Use of prophylactic doses of LWMH should be considered if the patient is unable to recommence warfarin in the immediate post-operative period.
- In some cases (very high risk of thromboembolism), patients may require post-operative bridging therapy with therapeutic LMWH, until the INR is at therapeutic level.
    - High bleeding risk procedures: Delay therapeutic LMWH until at least 48 h after surgery. Thromboprophylaxis should be given instead.
    - Low bleeding risk procedures: Restart therapeutic LMWH 6–12 h post-procedure, if haemostasis has been fully secured.

Recent Acute Coronary Syndrome and Dual Anti-Platelet Therapy:

- In patients with a recent acute coronary syndrome or coronary artery stent on dual antiplatelet therapy low bleeding risk procedures should proceed without interruption of antiplatelet therapy.
- In patients with a recent acute coronary syndrome or coronary artery stent on dual antiplatelet therapy, elective high bleeding risk procedures should, if possible be postponed in patients still requiring dual antiplatelet therapy. If surgery cannot be deferred, Aspirin should be continued and Clopidogrel or Ticagrelor interrupted from 5 days pre-op or Prasugrel from 7 days pre-op. (Confirmation with cardiology may be appropriate).

When to Stop Anti-Thrombotic Therapy Before Procedure

Use the Cockcroft-Gault Equation to convert serum creatinine to CrCL for DOACs:

$$\text{CrCl (ml/min)} = \frac{(140 - \text{Age (yrs)}) \times \text{weight (Kg)} \times \text{constant}}{\text{Serum Creatinine (µmol/L)}}$$

Constant = 1.23 for Males, 1.04 for females
CrCl can be calculated using MD+CALC Cockroft and Gault equation accessed via:
https://www.mdcalc.com/creatinine-clearance-cockcroft-gault-equation
It recognises the need to adjust for bodyweight in obese individuals and will calculate a modified estimate of CrCl with a range that is based on ideal bodyweight (IBW) to adjusted body weight (ABW).

NB *Always check the default units are correct when entering weight, serum creatinine and height. These can be changed using the drop down list if needed.*

Drug	Low Bleeding Risk	High Bleeding Risk	
Aspirin 75mg daily	N/A	N/A	Additional precautions
Clopidogrel	5- 7 days (7 if use of neuraxial anaesthesia)	5- 7 days (7 if use of neuraxial anaesthesia)	Consider discussion with cardiology/ Vascular if patient on dual antiplatelet/ anticoagulation in combination with antiplatelet.
Prasugrel	7days	7- days	
Ticagrelor	3 days (5 if use of neuraxial anaesthesia)	3 days (5 if use of neuraxial anaesthesia)	
Dipyridamole	Nil (24 hours if dual therapy)	Nil (24 hours if dual therapy)	
Unfractionated heparin S/C: prophylactic dose	4 hours	4 hours	
Unfractionated heparin IV: treatment dose	Stop infusion 4 hours before and check APTT	Stop infusion 4 hours before and check APTT	
LMWH: prophylactic dose	12 hours	12 hours	
LMWH: treatment dose	24 hours	24 hours	
Fondaparinux: prophylactic dose	24 hours	24 hours	
Warfarin and other VKAs	Surgeon discretion	5 days (INR ≤ 1.5)	
Dabigatran	CrCl ≥80ml/min: 24 hrs CrCl 50-80ml/min: 48 hrs CrCL 30-49ml/min:72 hrs	CrCl ≥80ml/min: 48 hrs CrCl 50-80ml/min: 72 hrs CrCL 30-49ml/min:96 hrs	
Rivaroxaban ≤10mg	12 hours	12 hours	
Rivaroxaban >10mg	CrCl ≥30ml/min: 24 hrs CrCl <30ml/min: 48 hrs	CrCl ≥30ml/min: 48 hours CrCl <30ml/min: 72 hours	
Apixaban	CrCl ≥30mL/min: 24 hrs CrCl <30mL/min: 48 hrs	CrCl ≥30mL/min: 48 hrs CrCl <30mL/min: 72 hrs	
Edoxaban	CrCl ≥30mL/min: 24hrs CrCl <30ml/min: 48 hrs	CrCl ≥30mL/min: 48 hrs CrCl <30ml/min: 72 hrs	

Time of First Dose after Neuraxial Procedure

N.B. P2Y12 inhibitors (Clopidogrel, Prasugrel & Ticagrelor), Warfarin, DOACs, should not be given whilst an epidural catheter is in situ.

Drug	Time of 1st dose after surgery, neuraxial procedure
Aspirin and NSAIDs	Nil
Clopidogrel	6 hours (or after epidural catheter removal)
Prasugrel	6 hours
Ticagrelor	6 hours
Dipyridamole	6 hours
Unfractionated heparin S/C: prophylactic dose	1 hour
Unfractionated heparin IV: treatment dose	4 hours
LMWH: prophylactic dose	4 hours
LMWH: treatment dose	4 hours (12 hours if bloody tap)
Fondaparinux: prophylactic dose	6 hours (12 hours after catheter removal)
Warfarin & other VKAs	After catheter removal
Dabigatran	6 hours (24 hours if bloody tap)
Rivaroxaban: prophylactic	6 hours (24 hours if bloody tap)
Rivaroxaban: treatment	6 hours (24 hours if bloody tap)
Apixaban	6 hours (24 hours if bloody tap)
Edoxaban	6 hours (24 hours if bloody tap)

Patients For Neck of Femur Surgery

- Delaying surgery because of anticoagulation should only be considered when the risk from bleeding outweighs the poorer survival associated with delay. See AAGBI Management of Hip Fractures Guideline (see below)
- The anti-platelet effect of NSAIDs will increase the bleeding risk in patients taking anticoagulants and are not advised in this patient group.
- Fascia iliaca blocks are relatively low risk of bleeding and, as such, may be considered in anticoagulated patients. Considerations should be made to bleeding risks and benefit from better analgesia, and a risk/ benefit decision made with the patient on a case by case basis.

Drug	Elimination half-life	Management	Acceptable to proceed with spinal
Aspirin	Irreversible effect on platelets	Proceed with surgery	Continue
Clopidogrel	Irreversible effect on platelets	Proceed with surgery under GA Monitor blood loss Consider platelet transfusion if concerns regarding bleeding	Yes, if GA poses greater risk to patient
Ticagrelor	8–12 h	Proceed with surgery with GA Monitor for blood loss Consider platelet transfusion if concerned about risk of bleeding	Yes, if GA poses greater risk to patient
Unfractionated i.v. heparin	1–2 h	Stop i.v. heparin 2–4 h pre-op	4 h
Low molecular weight heparin subcutaneous prophylactic dose	3–7 h	Last dose 12 h pre-op	12 h
Low molecular weight heparin subcutaneous **treatment dose**	3–7 h	Last dose 12–24 h pre-op. Monitor blood loss	24 h
Warfarin	4–5 days	5 mg vitamin K i.v. and repeat INR after 4 h Consider repeating. Consider prothrombin complex for immediate reversal	If INR < 1.5
Dabigatran	15–17 h	Consider surgery 24–48 h after last dose Review renal function Consider idarucizumab for immediate reversal	24–36 h if TT or dabigatran assay normal. If TT abnormal (prolonged) or assay > 50ng.ml^{-1}, give idarucizumab and proceed.
Rivaroxaban **Apixaban** **Edoxaban**	12 h	May be partially reversed with prothrombin complex. Consider surgery 12–24 h after last dose* Review renal function (*Consider at 12 h under general anaesthesia if bleeding risk acceptable)	24 h if CrCl ≥ 30 – no assay 48 h if CrCl < 30 - Or if available, proceed if DOAC specific assay ≤ 50 ng.ml^{-1} or reverse if DOAC specific assay > 50 ng.ml^{-1}

Emergent Procedures

Antiplatelet Agents

Urgent low bleeding risk surgery:

- Platelet transfusion should not be given.
- It is safe to operate early for patients on clopidogrel requiring Hip fracture surgery.

Urgent high-bleeding risk surgery:

- 2 pools of donor platelets may improve haemostasis if given at least two hours after the last dose of aspirin or 12–24 hours after the last dose of clopidogrel. Higher doses of platelets maybe required, within these time periods.
- Consider the use of pre-operative intravenous Tranexamic acid.

Warfarin and other VKAs:

- If surgery can wait for 6–8 hours, then administer phytomenadione (Vitamin K) 5mg iv to restore coagulation factors.
- If this is not possible, anticoagulation can be reversed with 25–50 u/kg of prothrombin complex concentrate (PCC) depending on the INR (Discuss with haematology).

DOACs:

Reversal agents:

- **Idarucizumab** (**Praxbind**) 5g iv bolus, should be used to reverse **Dabigatran** therapy prior to emergency invasive procedures where the bleeding risk is considered significant. Consultant Haematologist advice must be sought to approve use.
- **Andexanet (Andexxa)** should be used to reverse **Apixaban, Rivaroxaban or Edoxaban** (consult product literature for dosing).
- If Andexanet is unavailable, **Prothrombin Complex Concentrate (PCC)** can be considered for the reversal of **Apixaban**, **Edoxaban** or **Rivaroxaban** in life-threatening or uncontrolled bleeding or prior to emergency surgery [unlicensed use]. Consultant Haematologist advice must be sought to approve use of PCC.
- If a residual anticoagulant effect cannot be excluded, neuroaxial anaesthesia should ideally be avoided.
- Consider the use of Tranexamic acid to reduce bleeding in patients who have a residual anticoagulant effect.

Laboratory blood tests:

- A normal thrombin time (TT) can be interpreted as indicating that there is a minimal circulating concentration of Dabigatran.
- Normal prothrombin time (PT) and activated partial thromboplastin time (APTT) do not exclude significant concentrations of Dabigatran, Rivaroxaban or Apixaban.

References & Further Information

https://www.ukcpa-periophandbook.co.uk

Management of Hip Fractures 2020 Guideline, AAGBI

BCUHB Guideline: Management of NOACs in adults awaiting elective surgery (expiry July 2021)

Peri-operative management of anticoagulation & antiplatelet therapy. *British Journal of Haematology* 2016, 175, 602–613.

Regional anaesthesia and patients with abnormalities of coagulation. *AAGBI / RAUK Guidelines,* 2013.

Continuing Education in Anaesthesia Critical Care & Pain, Volume 12, Issue 1, 1 February 2012, Pages 11–16, https://doi.org/10.1093/bjaceaccp/mkr046 Published: 06 December 2011

Appendix 1: DOAC Quick Guide

Peri-Operative management of Direct Oral Anticoagulants (DOAC) in URGENT (not limb or life threatening) Surgery

Weigh patient and calculate Creatinine Clearance (CrCL) (nor estimated GFR)

Assess thrombotic and bleeding risks see guideline: YG CG 401

Pre-operative management of DOACs

CrCl (min/mL)	DABIGATRAN		RIVAROXIBAN & EDOXABAN		APIXABAN	
Bleeding risk	Standard risk and GA	High risk & Neuraxial Blocks	Standard risk and GA	High risk & Neuraxial Blocks	Standard risk and GA	High risk & Neuraxial Blocks
≥80	24 hrs	48 hrs	24hrs	48hrs	24 hrs	48 hrs
50-80	48 hrs	72 hrs				
30-49	72 hrs	96 hrs				
<30	Contraindicated		48 hrs	72hrs	48 hrs	72 hrs

Post -operative Management: when to restart DOAC post surgical intervention

Time of first dose of <u>any</u> DOAC after Neuraxial Procedure: 6 hours (24 hours if bloody tap)

Consider post operative risk of bleeding before re-starting.

All DOACs are rapidly absorbed and have a rapid onset of action, with peak anticoagulant activity at approximately 2-3 hours after oral ingestion. Attention to post-operative haemostasis is clinically important. Too early resumption of DOACs, especially within 24 hours of surgery, is associated with a two-to-fourfold increased risk of major bleeding.

Minor / Low Risk Procedures
Recommence 6 – 12 hours post-procedure if haemostasis has been fully secured.

High Risk Procedures / Increased Bleeding Risk
Do not recommence at full-dose until at least 48 hours post-procedure.

Consider prophylactic LMWH, commenced 6-12 hours post-op based on patient's thromboembolic risk and bleeding risk .LMWH should be discontinued immediately upon recommencing DOAC.

Reference:
Keeling D, Campbell Tait R, Watson H *et al* on behalf of the British Committee for Standards in Haematology. Peri-operative management of anticoagulation